Data Structures And Algorithms

FlowMax: Network Flow Optimization System

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Abstract

The FlowMax: Network Flow Optimization System project develops an efficient system for solving the Maximum Flow problem in a directed flow network. The system determines the maximum transferable flow from a source to a destination while respecting edge-capacity constraints and flow conservation rules. Edmonds-Karp, a BFS-based implementation of the Ford-Fulkerson method, is used to find augmenting paths and update the residual network. The system records the flow added at each step, displays the residual capacities, and identifies a minimum cut. The project demonstrates the practical application of graph algorithms, residual networks, optimization, and the Max-Flow Min-Cut theorem.

Introduction

The Maximum Flow problem is a fundamental network optimization problem in which the objective is to send the largest possible amount of flow from a source vertex to a destination or sink. Each directed edge has a limited capacity, and intermediate vertices must satisfy flow conservation. This project uses the Edmonds-Karp algorithm to repeatedly find augmenting paths and increase the total flow. A residual network is maintained after every augmentation, allowing the system to visualize how routing possibilities change until no further augmenting path exists.

Literature Survey

Various approaches have been developed to solve the Maximum Flow problem. The Ford-Fulkerson method uses augmenting paths to increase flow until no suitable path remains. Edmonds-Karp improves this approach by selecting augmenting paths using Breadth-First Search and provides a predictable polynomial-time complexity. Residual graphs are essential because they allow previously assigned flow to be reduced or rerouted through reverse edges. The Max-Flow Min-Cut theorem provides an optimality certificate by proving that the maximum flow value is equal to the capacity of a minimum cut.

Problem Statement and Objectives

The main problem addressed by this project is to develop an efficient network optimization system that determines the maximum flow between a source and destination. The system must respect edge capacities and maintain flow conservation at intermediate vertices. The primary objectives are to model a directed capacity network, implement Edmonds-Karp for maximum-flow computation, identify augmenting paths and bottleneck capacities, update the residual network after each augmentation, calculate the minimum cut, visualize the flow process, and demonstrate the Max-Flow Min-Cut theorem.

Results

The developed FlowMax system successfully calculates the maximum flow for the given network using Edmonds-Karp. The system identifies each augmenting path, calculates its bottleneck capacity, updates the forward and reverse residual capacities, and continues until no source-to-sink augmenting path remains. The final output includes the maximum flow value, flow assigned to network edges, the final residual network, and the minimum-cut capacity. The visualization makes the optimization process easier to understand and provides a direct verification of the Max-Flow Min-Cut theorem.

Hardware and Software requirements

The hardware requirements for implementing the project include a computer or laptop with an Intel Core i3 processor or above, a minimum of 4 GB RAM, at least 500 MB of available storage space, a keyboard, mouse, and monitor or display. The software requirements include Windows or Linux as the operating system, Java (JDK 8 or above) as the programming language, and an IDE or code editor such as Eclipse, IntelliJ IDEA, or Visual Studio Code. Standard Java data structures such as arrays, lists, queues, and hash maps can be used for graph and residual-network representation. Java Swing or JavaFX can be used for interactive network visualization.

Conclusion

The FlowMax project successfully implements the Maximum Flow problem using the Edmonds-Karp algorithm. The proposed approach models capacity-constrained directed networks, finds augmenting paths, maintains flow conservation, and updates residual capacities after each iteration. It determines the maximum transferable flow and identifies a corresponding minimum cut. The project demonstrates the practical application of graph algorithms, queues, residual networks, optimization, and the Max-Flow Min-Cut theorem. Overall, the system provides both an efficient solution and a clear visual explanation of the maximum-flow process.

Future Scope

The future scope of this project includes extending FlowMax to handle larger and more complex networks using optimized graph representations. The system can be enhanced with an interactive JavaFX or web-based interface for adding nodes, editing capacities, and selecting source and sink vertices. Animated augmenting paths and residual-network transitions can provide better visualization of the algorithm. Future versions can compare Ford-Fulkerson and Edmonds-Karp with other network optimization techniques and support real-world applications such as communication bandwidth allocation, traffic routing, pipeline distribution, logistics, and resource allocation.

Thank you